

Tutorial 2: Boolean Algebra

Question 1

Prove the following identities using Boolean algebra:

(a) $((A + B) (A'B'))' = A'B'$

$$\begin{aligned} & ((A+B).(A'B'))' \\ &= (A+B)' + (A'B') \\ &= (A'B') + (A'B') = A'B' \end{aligned}$$

(b) $(A + B')(A' + B' + C) = AC + B'$

$$\begin{aligned} & (A+B')(A'+B'+C) \\ &= (AA'+AB'+AC+A'B'+B'B'+B'C) \\ &= (AB'+A'B'+B'+B'C)+(AA'+AC) \quad (\text{Optional step}) \\ &= B' (A + A'+1+C) + AC \\ &= B'+AC \end{aligned}$$

Question 2

Simplify the following expression.

(a) $f(w,x,y,z) = x + xyz + x'yz + wx + w'x + x'y$

(b) $f(X,Y,Z) = (X'Y + XZ)(X + Y')$

Hint

(a) $x + y$

(b) XZ

$$\begin{aligned} \text{(a)} \quad f(w,x,y,z) &= x + xyz + x'yz + wx + w'x + x'y \\ &= x (1 + yz + w' + w) + yx'(1 + z) \\ &= x + x'y \quad \text{T5} \\ &= x + y \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad f(X,Y,Z) &= (X'Y + XZ)(X + Y') \\ &= X'YX + X'YY' + XZX + XZY' \\ &= 0 + 0 + XZ + XZY' \\ &= XZ + XZY' \\ &= XZ \end{aligned}$$

Question 3

Use Consensus Theorem to optimise the following expressions.

(a) $(x + y)z + x'y'w + zw$

(b) $(x' + y)wz + xy'v + vwz$

Hint

(a) $xz + yz + x'y'w$

(b) $x'wz + ywz + xy'v$

$$\begin{aligned} \text{(a)} \quad (x + y)z + x'y'w + zw &= (x' y')' z + x'y'w + zw \\ &= (x' y')' z + x'y'w \\ &= (x + y)z + x'y'w \\ &= xz + yz + x'y'w \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad (x' + y)wz + xy'v + vwz &= (xy')'wz + xy'v + vwz \\ &= (xy')'wz + xy'v \\ &= (x' + y)wz + xy'v \\ &= x'wz + ywz + xy'v \end{aligned}$$

Question 4

Given the function $f(x,y,z) = xy' + xz'$. Write f in the form of

(a) Canonical sum-of-product (SOP)

(b) Minterm list.

$$\begin{aligned} \text{(a)} \quad f(x,y,z) &= xy' + xz' \\ &= xy'(z' + z) + x(y' + y)z' \\ &= xy'z' + xy'z + xy'z' + xyz' \\ &= xy'z + xy'z' + xyz' \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad f(x,y,z) &= m_4 + m_5 + m_6 \\ &= \sum m(4,5,6) \end{aligned}$$

Question 5

Find the canonical SOP and minterm list for the logic functions below.

(a) $f(A,B) = A + B'$

$$\begin{aligned} A + B' &= AB + AB' + AB' + A'B' \\ &= AB + AB' + A'B' \quad (\text{Canonical SOP}) \\ &= m_3 + m_2 + m_0 \\ &= \sum m(0,2,3) \quad (\text{minterm list}) \end{aligned}$$

(b) $f(A,B,C,D) = A(B' + CD') + A'BC'D$

$$\begin{aligned} &A(B' + CD') + A'BC'D \\ &= AB' + ACD' + A'BC'D \\ &= AB'C'D' + AB'C'D + AB'CD' + AB'CD + ABCD' + A'BC'D \quad (\text{Canonical SOP}) \\ &= m_8 + m_9 + m_{10} + m_{11} + m_{14} + m_5 \\ &= \sum m(5,8,9,10,11,14) \quad (\text{minterm list}) \end{aligned}$$

Question 6

Construct the truth tables for the following switching functions.

(a) $f(A,B,C) = A(B + C')(B' + C)$

(b) $f(A,B,C,D) = A(B' + CD') + A'BC'D$

(a) Let $Y = f(A,B,C) = A(B + C')(B' + C)$
 $= ABC + AB'C'$

(b) Let $W = f(A,B,C,D) = A(B' + CD') + A'BC'D$
 $= AB' + ACD' + A'BC'D$

ABC	Y
000	0
001	0
010	0
011	0
100	1
101	0
110	0
111	1

ABCD	W	ABCD	W
0000	0	1000	1
0001	0	1001	1
0010	0	1010	1
0011	0	1011	1
0100	0	1100	0
0101	1	1101	0
0110	0	1110	1
0111	0	1111	0